

Microsoft India to provide free Internet using 'white space'

Microsoft India has plans to provide free last-mile Internet connectivity across India. It has proposed to use 'white space,' which is the unused spectrum between two TV channels, for this purpose.

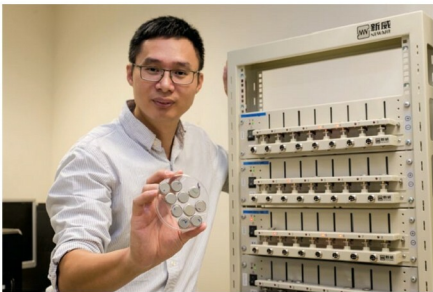
Bhaskar Pramanik, chairman, Microsoft India, said, "Wi-Fi has a range of only about 100 metres, whereas the 200-300MHz spectrum band available in the white space can reach up to 10km. This spectrum belongs mainly to Doordarshan and the government, and is not used at all. We have sought clearance for a pilot project in two districts."

He added, "The challenge is the lack of digital infrastructure across India. This initiative addresses this challenge in a cost-effective manner and creates an eco-system that will benefit everyone, including manufacturers of routers and other technology devices, and other technology companies, besides Microsoft."

If successful, the project can be quickly rolled out across the country and could give a boost to PM Modi's Digital India initiative, which proposes to use technology to deliver governance to every citizen of India, even in remote areas. The Digital India project aims to make all citizens digitally-empowered by providing broadband connectivity by connecting 250,000 Gram Panchayats via optic fibre cables.

Microsoft's initiative gives a push to Prime Minister's slogan of 'IT + IT = IT', which is Indian talent plus information technology equals India tomorrow and to the 'Make in India' campaign by encouraging the manufacture of equipment locally.

Microsoft had conducted widespread field trials on white space based Internet connectivity in Cambridge,



Using these ultra-fast charging batteries, cars can be charged in just a matter of minutes
(Courtesy: <http://www.overclock3d.net>)

US, in 2011. The technology has not been widely adopted anywhere in the world, but experts believe it can lead to a spurt in broadband connectivity in developing nations like India.

Battery that lasts 20 years

A team of engineers at Nanyang Technology University (NTU) has developed a battery that can be charged from 0 to 70 per cent within two minutes. This new innovation also hopes to save the pain of changing a battery every 4-5 years. These new batteries are designed to be energy-saving and these last for 20 years.

Researchers have used titanium dioxide, which is used in sunscreen lotions. Titanium dioxide lasts longer as compared to traditional graphite anode in batteries.

Researchers claim that these batteries will last for 10,000 charging cycles, whereas regular batteries last for only 500 charging cycles. Since these batteries have a much faster charging time, these can be used in electric vehicles too. A large prototype of this battery is currently being developed.

Keyboard software that can read emotions

Emotion is the only thing that differentiates humans from machines. Researchers at Islamic University of Technology have built a system intelligent enough to predict human emotions. They have designed a keyboard that reads your emotions using your keyboard usage. The software records your keyboard strokes and text to analyse what you were thinking while typing.

The software collects data input from the keyboard. The Java based module of the software collects fixed-text attributes, while C# based programs collect free text. The inventors integrated various models such as ISEAR, algebraic model with Jaccard to analyse the text input pattern. They tested the software using a group of volunteers, who typed phrases displayed on the screen. The software recorded and analysed the data to predict the emotional state of people when they typed those phrases. The test results showed 80 per cent accuracy in identifying their emotions.